Predicting Ski Jumps using State-Space Model

Živa Hegler and Neca Camlek

JSI (Jožef Stefan Institute)

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Outline

- Introduction
- Modeling Framework and Dataset
  - State-space models and Least squares method
  - Data processing
- Methodology
  - Different approaches and error function
- Main Results
  - Animation app
- Discussion and Conclusion

Motivation

- Planning of ski jump enlargement and anticipating risky conditions
- Goal: Predict ski jumps based on external factors

Figure: Flying hill in Planica

- State-space models (SSM)
- Least squares method (LS)

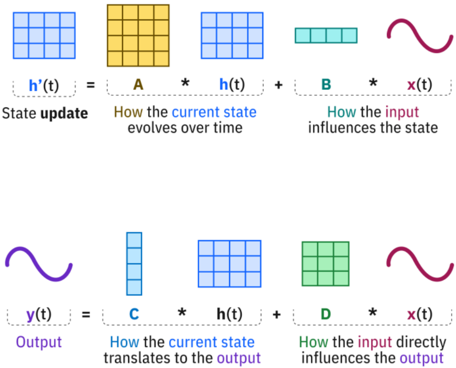


Figure: Schema of SSM

Data Description

- Cleaning the data
- Data structure
- Angles
- coordinate system on a main ski jumping hill in Planica

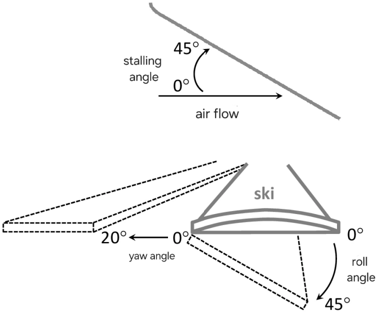


Figure: Representation of different measured angles.

Ski jumping hill

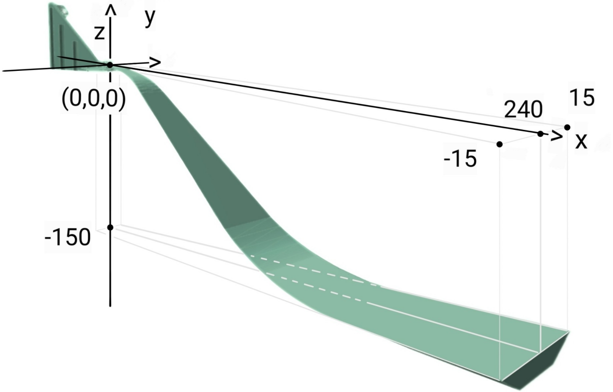


Figure: Ski jumping hill with a coordinate system placed on top.

Different approaches and error function

Different approaches:

- pure SSM
- SSM + PINN (Physics Informed Neural Networks)

Error function:

- Error function
- leave-one-out cross-validation

different add-ons to the model:

- all 12 wind sensors or only 3 wind zones
- squared wind values

Flight simulation example

Results:

- train error 1.76 m
- test error 1.84 m.

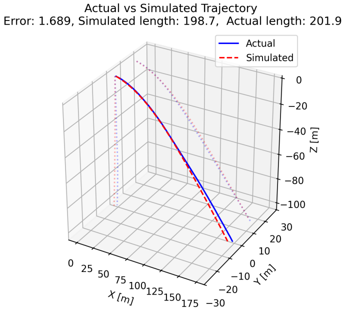


Figure: Graph with a simulated and actual flight trajectory.

Application

- Description of the application
- Observing measured and predicted jumps

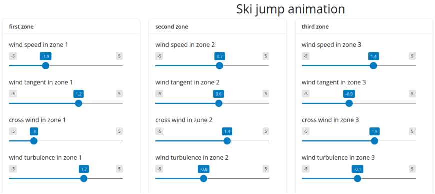
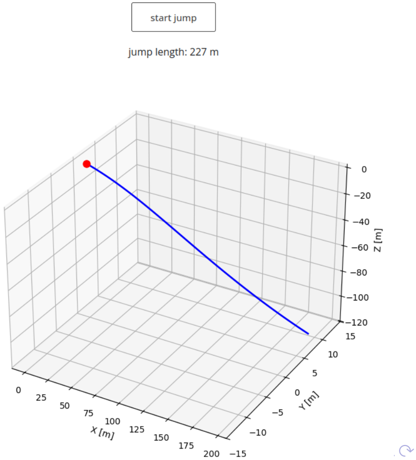


Figure: Adjusted sliders in the app



Interpretation

- Limitations
- Future work and potential improvements
- Short recap

Thank you!